

INFO8006 Introduction to Artificial Intelligence

Exercise session 3: Quantifying uncertainty

RECAP

Definitions

A probability is a degree of belief in a proposition.

Sample space	Ω , of possible worlds ω
Axioms	$0 \leq P(\omega) \leq 1$, $\sum_{\omega \in \Omega} P(\omega) = 1$
Event	$A \subseteq \Omega$, $P(A) = \sum_{\omega \in A} P(\omega)$
Random variable	$X : \Omega \rightarrow D_X$, with $X = x$ the event $\{\omega \in \Omega : X(\omega) = x\}$
Distribution	$\mathbf{P}(X)$, the numbers $P(x) = P(X = x)$ for $x \in D_X$
Joint distribution	$\mathbf{P}(X_1, \dots, X_n)$, the numbers $P(x_1, \dots, x_n)$
Conditional probability	$P(a \mid b) = P(a, b) / P(b)$, for $P(b) > 0$
Conditional distribution	$\mathbf{P}(X \mid Y)$, one $\mathbf{P}(X \mid y)$ for each $y \in D_Y$

Random variables are written in upper case and their values in lower case; a boolean variable A takes the values a and $\neg a$. $P(x)$ is a number, $\mathbf{P}(X)$ a table.

Rules

Complement	$P(\neg a) = 1 - P(a)$
Union	$P(a \vee b) = P(a) + P(b) - P(a \wedge b)$
Product rule	$P(a, b) = P(a \mid b)P(b) = P(b \mid a)P(a)$
Sum rule, total probability	$P(x) = \sum_y P(x, y) = \sum_y P(x \mid y)P(y)$
Chain rule	$P(x_{1:n}) = \prod_i P(x_i \mid x_{1:i-1})$
Bayes' rule	$P(a \mid b) = P(b \mid a)P(a) / P(b)$
Independence, $X \perp Y$	$P(x, y) = P(x)P(y)$
Conditional independence, $X \perp Y \mid Z$	$P(x, y \mid z) = P(x \mid z)P(y \mid z)$
Naive Bayes	$P(c, e_{1:n}) = P(c) \prod_i P(e_i \mid c)$

The rules hold for all values, and between distributions. Naive Bayes assumes the effects $e_{1:n}$ conditionally independent given the cause c . The joint distribution answers every query: select the entries consistent with the evidence, sum out the other variables, normalize.

Bayes' rule

A model is given in the *causal* direction, a *prior* $\mathbf{P}(A)$ and a *likelihood* $\mathbf{P}(B \mid A)$. The *diagnostic* question is answered by the *posterior* $\mathbf{P}(A \mid b) = \frac{1}{Z} \mathbf{P}(b \mid A) \mathbf{P}(A)$, normalized by the probability of the *evidence*, $Z = P(b) = \sum_a P(b \mid a)P(a)$.

In session: exercises 1, 2, 3, 5.

Exercise 1 Rational beliefs (AIMA ch. 12)

Let $A, B \subseteq \Omega$ be two *events*. An agent holds the beliefs $P(A) = 0.4$ and $P(B) = 0.3$. What ranges of probabilities would it be rational for the agent to hold for the events $A \cup B$ and $A \cap B$? What if the agent also believes that $P(A | B) = 0.5$?

Solution.

1. Since $P(E) = \sum_{\omega \in E} P(\omega)$ sums nonnegative terms, $E \subseteq F$ implies $P(E) \leq P(F)$. With $A, B \subseteq A \cup B$ and $A \cap B \subseteq A, B$, and the union rule $P(A \cup B) = P(A) + P(B) - P(A \cap B)$, the agent's beliefs should satisfy

$$\begin{aligned} 0.4 &= \max\{P(A), P(B)\} \leq P(A \cup B) \leq \min\{1, P(A) + P(B)\} = 0.7 \\ 0 &= \max\{0, P(A) + P(B) - 1\} \leq P(A \cap B) \leq \min\{P(A), P(B)\} = 0.3. \end{aligned}$$

The two beliefs are not independent choices: by the union rule, $P(A \cup B) + P(A \cap B) = 0.7$. The extremes are $B \subseteq A$, with $P(A \cup B) = 0.4$ and $P(A \cap B) = 0.3$, and disjoint events, with 0.7 and 0.

2. If the agent believes that $P(A | B) = 0.5$, it should also believe that

$$P(A \cap B) = P(A | B)P(B) = 0.5 \times 0.3 = 0.15,$$

by the product rule. Then, the agent should rationally believe that

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.4 + 0.3 - 0.15 = 0.55.$$

Exercise 2 Full joint distribution (AIMA ch. 12)

A monitoring program supervises the bearings of a wind turbine. A worn bearing (*fault*) tends to make the turbine vibrate (*vibration*) and to trigger the alarm of a diagnostic probe (*alarm*). The program holds the joint distribution $\mathbf{P}(V, A, F)$ below, over the boolean random variables V (vibration), A (alarm) and F (fault). Compute the following probabilities.

	v		$\neg v$	
	a	$\neg a$	a	$\neg a$
f	0.108	0.012	0.072	0.008
$\neg f$	0.016	0.064	0.144	0.576

- (a) $P(v)$

Solution.

$$P(v) = \sum_{a'} \sum_{f'} P(v, a', f') = 0.108 + 0.012 + 0.016 + 0.064 = 0.2$$

- (b) $P(f)$

Solution.

$$P(f) = \sum_{v'} \sum_{a'} P(v', a', f) = 0.108 + 0.012 + 0.072 + 0.008 = 0.2$$

(c) $P(v \mid f)$

Solution.

$$P(v \mid f) = \frac{P(v, f)}{P(f)} = \frac{\sum_{a'} P(v, a', f)}{P(f)} = \frac{0.108 + 0.012}{0.2} = 0.6$$

(d) $P(f \mid v \vee a)$

Solution.

$$\begin{aligned} P(f \mid v \vee a) &= \frac{P(f, v \vee a)}{P(v \vee a)} = \frac{P(f) - P(\neg v, \neg a, f)}{1 - P(\neg v, \neg a)} \\ &= \frac{0.2 - 0.008}{1 - 0.584} = 0.4615 \end{aligned}$$

Exercise 3 Fraud detection (AIMA ch. 12)

A fraud detection program screens card payments. It is 99% accurate, *i.e.* it flags a fraudulent payment with probability 0.99 and a legitimate payment with probability 0.01. Fraud is rare: it strikes only 1 payment in 10 000. The program has just flagged your payment. Why is it good news, for you, that fraud is rare? What is the probability that the payment is actually fraudulent?

Solution. Let f be the event that the payment is fraudulent and a the event that it is flagged. Rarity is good news because the prior $P(f)$ is small: most flagged payments are legitimate payments flagged by mistake. Indeed,

$$\begin{aligned} P(f \mid a) &= \frac{P(a \mid f)P(f)}{P(a)} \\ &= \frac{P(a \mid f)P(f)}{P(a \mid f)P(f) + P(a \mid \neg f)P(\neg f)} \\ &= \frac{0.99 \times 10^{-4}}{0.99 \times 10^{-4} + 0.01 \times (1 - 10^{-4})} = 0.0098. \end{aligned}$$

Exercise 4 Conditional independence (AIMA ch. 12)

For each of the following statements, either prove it is valid or give a counterexample.

(a) If $P(a, b, c) > 0$ and $P(a \mid b, c) = P(b \mid a, c)$, then $P(a \mid c) = P(b \mid c)$.

Solution. True. By the product rule, $P(a \mid b, c)P(b, c) = P(a, b, c) = P(b \mid a, c)P(a, c)$. Since $P(a, b, c) > 0$, $P(a \mid b, c) = P(b \mid a, c) > 0$, and dividing by it gives $P(b, c) = P(a, c)$ and thereby $P(b \mid c) = P(a \mid c)$.

(b) If $P(a \mid b, c) = P(a)$, then $P(b \mid c) = P(b)$.

Solution. False. The statement $P(a \mid b, c) = P(a)$ merely states that a is independent of b and c , it makes no claim regarding the dependence of b and c . Counterexample: a and b record the results of two independent coin flips, and $c = b$.

(c) If $P(a \mid b) = P(a)$, then $P(a \mid b, c) = P(a \mid c)$.

Solution. False. While the statement $P(a \mid b) = P(a)$ implies that a is independent of b , it does not imply that a is conditionally independent of b given c . Counterexample: a and b record the results of two independent coin flips, and $c = a \vee b$.

Exercise 5 Driving styles (AIMA ch. 13)

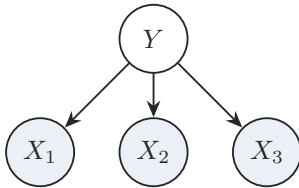
A self-driving car shares the highway with another car, whose driving style it does not know. It models three styles, cautious (a), normal (b) and aggressive (c), equally likely a priori, under which the other car changes lanes during a given minute with probabilities 0.2, 0.6 and 0.8, respectively. The self-driving car observes the other car for three minutes, recording whether it changes lanes (l) or keeps its lane (k) during each minute, as X_1 , X_2 and X_3 .

- (a) Draw the Bayesian network corresponding to this setup and define the corresponding conditional probability tables (CPTs).

Solution. The process is described with the following random variables:

- $Y \in \{a, b, c\}$, the driving style of the other car;
- $X_i \in \{l, k\}$, its behaviour during the i -th minute.

The style Y is the parent of each X_i : the observations are conditionally independent given the style.



y	$P(y)$	$P(X_i = l \mid y)$
a	$1/3$	0.2
b	$1/3$	0.6
c	$1/3$	0.8

- (b) Determine the most likely driving style if the other car changed lanes during the first two minutes and kept its lane during the third.

Solution. We compute the posterior $\mathbf{P}(Y \mid X_1 = l, X_2 = l, X_3 = k)$. By Bayes' rule,

$$\mathbf{P}(Y \mid X_1 = l, X_2 = l, X_3 = k) = \frac{1}{Z} \mathbf{P}(X_1 = l, X_2 = l, X_3 = k \mid Y) \mathbf{P}(Y).$$

Since X_1 , X_2 and X_3 are conditionally independent given Y ,

$$\mathbf{P}(X_1, X_2, X_3 \mid Y) = \mathbf{P}(X_1 \mid Y) \mathbf{P}(X_2 \mid Y) \mathbf{P}(X_3 \mid Y).$$

Then,

$$P(y \mid X_1 = l, X_2 = l, X_3 = k) = \frac{1}{Z} \times \frac{1}{3} \times \begin{cases} 0.2 \times 0.2 \times 0.8 = 0.032 & \text{if } y = a \\ 0.6 \times 0.6 \times 0.4 = 0.144 & \text{if } y = b \\ 0.8 \times 0.8 \times 0.2 = 0.128 & \text{if } y = c, \end{cases}$$

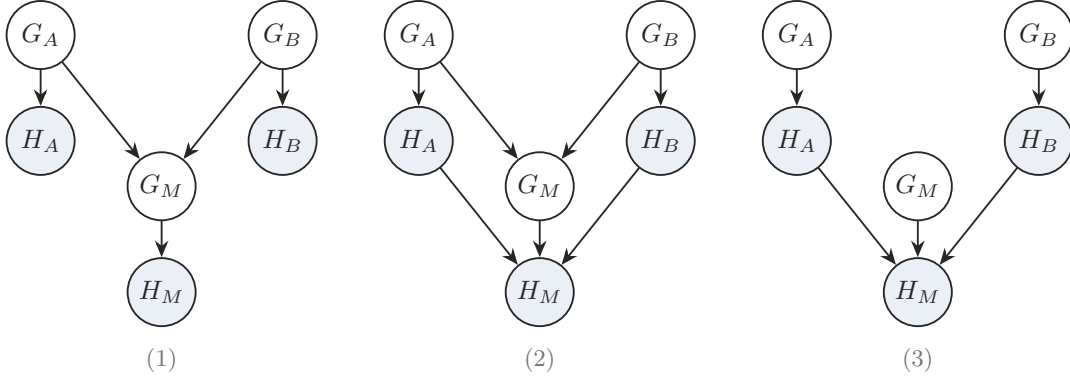
with $Z = \frac{1}{3}(0.032 + 0.144 + 0.128)$. The posterior is $\langle 0.105, 0.474, 0.421 \rangle$: the normal style is the most likely, though the aggressive one is nearly as probable. Finding the most likely style only needs the unnormalized values. The order of the observations does not matter: any two lane changes and one lane keep have the same probability given each style.

Exercise 6 Evolutionary model merging (AIMA ch. 13)

Evolutionary model merging builds a new language model M from two parent models A and B : each component of M is copied from one of the parents. Consider one component, and let G_x be the language it was mostly trained on in model x , with possible values e (English) or f (French). A common *hypothesis* is that the default language of a model follows from this component by a simple mechanism: the language H_x in which model x answers by default is the same as G_x with some probability s .

Furthermore, M copies the component from either parent with equal probability, and a random perturbation flips it with a small nonzero probability m .

- (a) Which of the following networks claim that $\mathbf{P}(G_M | G_A, G_B) = \mathbf{P}(G_M)$? Shaded variables are observed.



Solution. The third one, as the node G_M does not have any parents.

- (b) Which of the networks make independence claims that are consistent with the hypothesis about the default language?

Solution. The first and second ones. The second one is also valid as it makes *fewer* assumptions about the dependencies between the variables, *i.e.* it is more general than the first one.

- (c) Write down the CPT for the G_M node in the first network, in terms of s and m .

Solution.

g_A	g_B	$P(G_M = f g_A, g_B)$
f	f	$1 - m$
f	e	0.5
e	f	0.5
e	e	m

The CPT does not depend on s , which only relates G_M and H_M .

- (d) Suppose that $P(G_A = f) = P(G_B = f) = q$. In the first network, derive an expression for $P(G_M = f)$ in terms of s , m and q .

Solution.

$$\begin{aligned}
 P(G_M = f) &= \sum_{g_A} \sum_{g_B} P(G_M = f, g_A, g_B) \\
 &= \sum_{g_A} \sum_{g_B} P(G_M = f | g_A, g_B) P(g_A) P(g_B) \\
 &= (1 - m)q^2 + 0.5(1 - q)q + 0.5q(1 - q) + m(1 - q)^2 \\
 &= q + m - 2qm
 \end{aligned}$$

Again, s does not appear.

- (e) Over many generations of merges, we expect the distribution of the component to be the same in the parents and in the merged models. Use this to calculate the value of q . Only about 10 % of the merged models answer in French by default: explain why the hypothesis described at the beginning of this question must be wrong.

Solution. At equilibrium, $P(G_M = f) = P(G_A = f) = q$, so $q = q + m - 2qm$, *i.e.* $m(1 - 2q) = 0$. Since $m > 0$, $q = 0.5$. Then $P(H_M = f) = sq + (1 - s)(1 - q) = 0.5$ whatever s : half of the merged models should answer in French, against about 10 % observed, which invalidates the hypothesis.

1. The Bayes' rule states that ...

- ☐ $P(x | y) = \frac{P(y | x)P(y)}{P(x)}.$
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- ☐ $P(x) = \frac{P(x | y)P(y)}{P(x)}.$

2. Which of the following is true?

- ☐ $P(x, y | z) = \frac{P(x | y, z)}{P(y, z | x)}.$
- ☐ $P(x, y, z) = P(x)P(y)P(z).$
- ☒ $P(x, y | z) = P(y | x, z)P(x | z).$
- ☐ $P(x | y, z) = \frac{P(x, y, z)}{P(y)P(z)}.$

3. Naive Bayes model assumes ...

- ☒ conditional independence between the effects, given the cause.
- ☐ independence between the effects.
- ☐ independence between the effects and the cause.
- ☐ conditional independence between each effect and the cause, given the other effects.

4. The Bayes' rule can be read as "The posterior is equal to ...

- ☐ the product of the likelihood and the prior".
- ☐ the ratio between the posterior distribution and the evidence, times the prior".
- ☐ the likelihood to prior ratio multiplied by the evidence".
- ☒ the likelihood to evidence ratio multiplied by the prior".